Project Title: Predictive Exit Interview System

Prepared For: HR & Executive Management

1. Executive Summary & Objective Employee turnover incurs significant costs in recruitment, onboarding, and lost productivity. The objective of this project is to develop a machine learning framework that proactively identifies employees at high risk of attrition *before* they resign. This allows HR to conduct "stay interviews" and deploy targeted retention strategies.

2. Scope & Data Source

- **In Scope:** Developing a binary classification model (Attrition: Yes/No) using historical employee data.
- **Out of Scope:** Automating the actual HR interventions or compensation adjustments.
- **Data Source:** IBM HR Analytics Attrition Dataset (35 features, ~1,470 records).

3. Success Criteria (Metrics)

- **Primary Metric: Recall (Sensitivity)** for the "Yes" class must be > 80%. It is better for the model to accidentally flag a loyal employee as a flight risk (False Positive) than to miss an employee who is actually going to quit (False Negative).
- **Secondary Metric:** F1-Score to ensure precision doesn't drop too low.

4. Key Stakeholders

- **HR Business Partners:** Primary users of the dashboard.
- **Department Managers:** Implement retention strategies.
- **Data Science Team:** Model development, deployment, and monitoring.

5. Risks & Constraints

- **Data Privacy:** The dataset contains sensitive employee metrics (income, performance ratings) requiring strict access controls.
- **Model Bias:** Historical biases in promotions or pay could skew predictions.